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Neurophysiological evidence for the validity of verbal strategy reports in mental arithmetic

Roland H. Grabner[a,*,1], Bert De Smedt[b,1]

[a] Institute for Behavioral Sciences, Swiss Federal Institute of Technology (ETH) Zurich, Universitätsstrasse 41, UNO C15, CH-8092 Zurich, Switzerland
[b] Department of Educational Sciences, Katholieke Universiteit Leuven, Andreas Vesaliusstraat 2, Leuven, Belgium

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ABSTRACT

Behavioral research has shown that arithmetic problems (e.g., 6 + 2=) are solved with various strategies, which can be inferred from the size of the presented problems or from trial-by-trial verbal strategy reports. The validity of these verbal strategy reports, however, has been repeatedly questioned. In the present electroencephalography study, we compared the association of both approaches with the oscillatory brain responses during arithmetic problem solving. Nineteen adults solved small and large addition and subtraction problems and indicated the applied strategy (fact retrieval vs. procedure use) on a trial-by-trial basis by means of verbal strategy reports. Analysis of event-related (de-)synchronization (ERS/ERD) in theta and alpha frequencies revealed a general convergence of verbal strategy reports and the problem size approach, with fact retrieval being accompanied by higher left-hemispheric theta ERS, and procedural strategies being reflected in higher widespread ERD in the lower alpha band and bilateral parietooccipital ERD in the upper alpha band. A direct comparison of the neurophysiological data from both approaches suggests a higher sensitivity of verbal strategy reports to problem solving strategies applied in mental arithmetic, particularly for large problems. Taken together, the current data provide the first neurophysiological evidence for the validity of verbal strategy reports.

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1. Introduction

Whenever people are performing arithmetic (e.g., 6 + 9=), they use different strategies (Campbell and Xue, 2001; LeFevre et al., 1996; Siegler et al., 1996). They solve these problems either by directly retrieving the answer from long-term memory (retrieval strategy), or they use a more time-consuming procedural strategy, such as counting or transforming the problem into smaller subproblems to arrive at the solution (e.g., 8 + 6 = is solved by 8 + 2 = 10 and 10 + 4 = 14). In skilled adults, retrieval is typically more often used in problems with smaller operands (i.e., sums ≤10 or small problems) than in those with larger operands (i.e., sums >10 or large problems), whereas procedural strategies occur more often in large than in small problems (Campbell and Xue, 2001; LeFevre et al., 1996). This so-called effect of problem size (Zbrodoff and Logan, 2005) has been used in neuroimaging and neurophysiological studies to elucidate the brain correlates of arithmetic strategy use by contrasting brain responses of small vs. large problem sizes (e.g., De Smedt et al., 2009; Grabner et al., 2007a; Jost et al., 2004a,b; Kong et al., 2005; Ku et al., 2010; Nunez-Pena et al., 2006; Stanescu-Cosson et al., 2000; Zago et al., 2001). An important limitation of this approach is the assumption that all problems of a particular size are solved by one specific strategy, which might result in misleading conclusions on how people solve these problems (Siegler, 1987). Indeed, many behavioral studies that used concurrent verbal strategy reports – i.e., participants had to indicate on a trial-by-trial basis how they solved each problem – suggest that a variety of strategies is applied in problems of a given size (Campbell and Xue, 2001; LeFevre et al., 1996). Moreover, the distribution of strategies even differs across operations (Campbell and Xue, 2001; Seyler et al., 2003), and between cultures (Campbell and Xue, 2001). Verbal strategy reports are thus another, more direct way to capture individuals' strategy use. The validity of these reports, however, has been controversially discussed (Kirk and Ashcraft, 2001; Smith-Chant and LeFevre, 2003), and there exists no neuroimaging or neurophysiological study that has used concurrent verbal strategy reports to infer the brain correlates of arithmetic strategies. In the present high-resolution electroencephalography (EEG) study, we collected verbal strategy reports on a trial-by-trial basis in problems of different sizes and investigated the validity of verbal strategy reports and the problem size approach by evaluating their association with neurophysiological data.

* Corresponding author. Tel.: +41 44 632 50 53; fax: +41 44 632 12 19.
E-mail address: grabner@ifv.gess.ethz.ch (R.H. Grabner).
[1] Both authors equally contributed to this study and should be regarded as joint first-authors.

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Functional magnetic resonance imaging (fMRI) studies have revealed that solving small problems is associated with stronger activation in left-hemispheric perisylvian language regions including the supramarginal and angular gyri, presumably reflecting the retrieval of verbally stored arithmetic facts from memory. Large problems, on the other hand, were found to more strongly activate a wide-spread network of prefrontal and parietal regions bilaterally including the intraparietal sulcus (Grabner et al., 2007a; Kong et al., 2005; Stanescu-Cosson et al., 2000; Zago et al., 2001). The latter region has been interpreted to reflect the involvement of quantity manipulations during calculation (Ansari, 2008; Dehaene et al., 2003). In contrast to the fMRI studies, which elucidated the brain areas engaged during mental arithmetic, EEG studies have mainly focused on the time-course of brain activity during arithmetic problem solving by investigating event-related potentials (ERPs). These studies have shown that problem size modulates the amplitude of late ERP components occurring after about 300–400 ms (e.g., Jost et al., 2004a,b; Ku et al., 2010; Nunez-Pena et al., 2006), with a more pronounced negativity for large than for small problems over right temporo-parietal cortices.

In addition to the time-course of neurophysiological responses, there is growing interest in how the task-related neuronal networks are formed and interact with each other. A well-established way to investigate the dynamics of functional network formation during cognitive processing is the analysis of induced EEG activity. Different from ERPs (evoked activity), which reflect the (time- and phase-locked) responses of the brain to a single event, induced activity refers to the (time-locked) change in oscillatory EEG activity elicited by an event or task. These changes are related to the coupling and uncoupling of functional networks in the brain (Bastiaansen et al., 2006; Klimesch et al., 2005; Neuper and Pfurtscheller, 2001) and are frequently quantified in terms of event-related synchronization (ERS) and desynchronization (ERD; for a review, cf. Neuper and Klimesch, 2006). Specifically, the ERS/ERD method calculates the percentage amount of band power increases (ERS) or decreases (ERD) in a frequency band from a pre-stimulus reference interval to an activation interval (for a more detailed description of this method, see Pfurtscheller and Lopes da Silva, 2005).

Studies applying the ERS/ERD methodology have accumulated a large basis of evidence suggesting a differential functional significance of different frequency bands. Increases in theta activity (about 3–6 Hz) have been associated with memory encoding and retrieval, in general (Burgess and Gruzelier, 2000; Jensen and Tesche, 2002; Klimesch et al., 1997), and with retrieval of lexical-semantic information from long-term memory, in particular (Bastiaansen et al., 2006; Bastiaansen et al., 2005; Grabner et al., 2007b). Decreases in alpha activity (about 8–13 Hz) have been observed in response to various cognitive task demands and have been found to be related to task difficulty (Ku et al., 2010; Neubauer et al., 2006). Therefore, alpha ERD has been interpreted as an index of general cortical activation or invested cognitive resources (Pfurtscheller and Lopes da Silva, 1999).

In the domain of mental arithmetic few studies have investigated oscillatory EEG activity. Earle et al. (1996) reported higher left-hemispheric theta bandpower when participants solved arithmetic problems compared to inserting an arithmetic sign into an equation, which was interpreted to reflect fact retrieval during arithmetic problem solving. Harmony et al. (1999) found task-related activation changes during a complex arithmetic task in the theta (bandpower increases) and in the alpha frequency range (bandpower decreases). Theta effects were interpreted to indicate sustained attention, whereas the alpha effect was interpreted as memory retrieval. Recently, De Smedt et al. (2009) used the ERS/ERD method in a high-resolution EEG study to investigate the brain responses during addition and subtraction. They presented small (sums ≤ 10) and large (sums > 10) problems, which were designed to elicit the use of retrieval and procedural strategies, respectively. Findings revealed a strong dissociation between ERS/ERD patterns in the theta and alpha frequency bands for both problem sizes: higher left-hemispheric theta ERS was observed for small problems whereas large problems elicited stronger bilateral alpha ERD. In both bands, the differences were mainly located in the parietooccipital area, which may cover task-related parietal brain regions including the angular gyrus and the intraparietal sulcus. Against the background of the functional significance of the theta band, reflecting the retrieval of lexico-semantic information, and the alpha band, indicating the amount of cognitive resources invested in a task, the authors concluded that the pronounced left-hemispheric theta ERS in small problems indexed fact retrieval processes whereas the strong bilateral alpha ERD in large problems reflected the higher cognitive investment during arithmetic procedures. Similar ERS/ERD data in the theta and alpha band were recently reported by Moeller et al. (2010).

To the best of our knowledge, there exists only one brain imaging study that has used verbal strategy reports, instead of comparing small and large problems, to investigate the neural correlates of retrieval and procedural strategies. Grabner et al. (2009a) presented participants with arithmetic problems of all four operations during fMRI. After the scanning, participants had to solve half of the problems again in a paper-and-pencil test and had to indicate after every problem which strategy they used. Brain activity during the solution of problems for which the participants indicated use of retrieval vs. procedure use was compared. It turned out that retrieval strategy use was accompanied by stronger activation in the left angular gyrus, whereas problems for which participants indicated the use of a procedural strategy were associated with stronger activation in a widespread network of frontal and parietal regions, bilaterally.

It should be noted that the strategy assessment of Grabner et al. (2009a) differs from the one that is typically employed in behavioral studies (Siegler, 1987). Grabner et al. (2009a) did not apply an online or concurrent assessment (i.e., immediately after solving each problem in the scanner) on a trial-by-trial basis, which complicates the interpretation of the brain activation data. In fact, an online trial-by-trial strategy assessment is difficult to implement in an fMRI design, because the sluggish hemodynamic response does not easily allow one to analyze the problem solving phase independently from the strategy selection phase. In this context, the high temporal resolution of EEG in the range of milliseconds offers a strong methodological advantage for such a design. Apart from a clear separability of brain activity related to problem solving and strategy selection, the time periods of problem solving can be defined for each trial and for each individual with high temporal accuracy (e.g., De Smedt et al., 2009). In this vein, the EEG responses related to problem solving using different strategies can be isolated, and differences in response latencies between strategies and participants can be controlled.

The present high-resolution EEG study is the first in which the application of arithmetic strategies was assessed using concurrent trial-by-trial verbal strategy reports. Similar to the study by De Smedt et al. (2009), we presented participants with addition and subtraction problems of small and large problem size, with small problems having sums or minuends ≤ 10 and large problems having sums or minuends between 11 and 37. After solving each problem, participants were required to indicate whether they solved the problem using fact retrieval, the application of a procedure or any other strategy.

By using neurophysiological data, we sought to investigate the validity of verbal strategy self-reports and, on the other hand, of the problem size approach to detect arithmetic problem solving strategies. To this end, we examined the EEG data in two ways. First, we

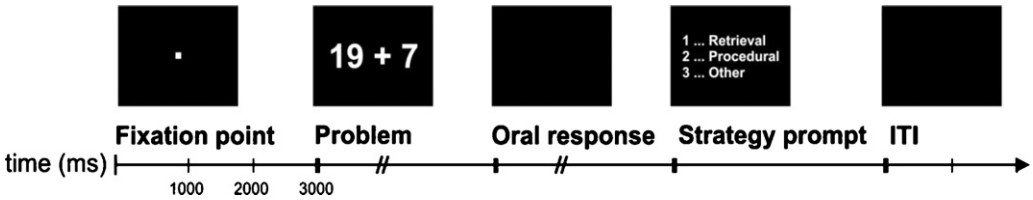

Fig. 1. Schematic display of one EEG trial. ITI = inter-trial interval of 2000 ms length.

analyzed the ERS/ERD patterns of problems solved by fact retrieval and those solved by procedural strategies to investigate whether these patterns generally converge with those obtained on the basis of problem size. Second, we directly compared the verbal strategy report and problem size approaches by evaluating (a) whether the ERS/ERD patterns differ between reported strategies when the problem size is held constant, and (b) whether ERS/ERD patterns differ between problem sizes when the reported strategy is held constant. If ERS/ERD patterns differ between reported strategies when the problem size is held constant, then verbal strategy reports are more sensitive to actual strategy use than the problem size method. If ERS/ERD patterns differ between problem sizes when the reported strategy is held constant, then problem size rather than verbal strategy report data is related to the use of different strategies.

Similar to De Smedt et al. (2009), we restricted our analysis to the theta and alpha frequency bands, which have been found to be sensitive to arithmetic processes, but additionally distinguished lower and upper alpha frequencies (cf. Klimesch, 1999). In general, we hypothesized that retrieval and small problems would be associated with larger left-hemispheric theta ERS than procedural problems and large problems. In the lower alpha band, which is topographically less differentiated than the upper alpha band (Klimesch, 1999), we expected higher overall ERD for procedural and large problems compared to retrieval and small problems. In the upper alpha band, a similar result pattern is expected but the difference should be especially pronounced over task-related parietooccipital cortices bilaterally.

2. Methods

2.1. Participants

Twenty-five right-handed healthy university students without known mathematical difficulties participated in this study. Six participants were excluded from analysis. Three of them were excluded due to technical problems during EEG acquisition and to loss of data. Two participants were further removed from the dataset because they indicated on more than half of the trials to have used a strategy other than retrieval or procedure use. Another participant needed to be excluded due to the lack of valid (artifact-free and correctly solved) EEG trials for analyzing retrieval strategy use in large problems. The remaining sample of 19 participants comprised 10 males and 9 females, aged between 20 and 31 years (M = 24.47, SD = 2.93 years). The study was approved by the local ethics committee (Swiss Federal Institute of Technology Zurich, Switzerland). All participants gave written informed consent.

2.2. Materials and apparatus

Participants solved 84 addition and 84 subtraction problems of different problem size. For each arithmetic operation, 36 small and 48 large problems were administered. No zero-, one-, or tie problems (e.g., 6 + 6) were included. Small addition problems had sums below or equal to 10 (e.g., 3 + 5); all possible 24 small problems were included and half of them were presented two times. Large addition problems all required carrying and had sums between 11 and 37 (e.g., 19 + 7), including one-digit + one-digit problems (e.g., 4 + 8), two-digit + one-digit problems (e.g., 28 + 3) and two-digit + two-digit problems (e.g., 14 + 19). For each decade (i.e., 11–19, 21–29, 31–39) we selected 16 problems from all the possible carry problems within that decade. Notably, our category of large problems included two-digit problems that are larger in problem size than is typically the case in many of the existing studies which focused on single-digit arithmetic (e.g., Campbell and Xue, 2001). This inclusion of two-digit problems was needed to ensure a sufficient number of trials on which procedural strategies were reported. The position of the largest addend was counterbalanced for both problem types. The subtraction problems were the exact inverse of the addition problems. A list of all presented problems is included in Table S1 of the supplementary online material.

The temporal sequence of one EEG trial is depicted in Fig. 1. Each trial started with the presentation of a fixation point for 3000 ms, followed by the arithmetic problem. The participant was instructed to solve the problem as accurately and quickly as possible and to speak the solution into a voice-activated microphone (voice-key) that was connected to the computer. The time period between problem onset and speech onset (as indicated by the voice key) represented the response latency. The experimenter entered the oral response (i.e., the calculation result) into the computer. These oral responses were also digitally recorded and cross-checked with the manual response of the experimenter after the test session. A timeout of 7000 ms was applied. Immediately following the response, a strategy prompt appeared in the center of the screen asking participants whether they solved the problem by (a) fact retrieval (e.g., remembering the solution or knowing the solution by heart), (b) application of a procedure (e.g., transformation of the problem or counting) or (c) any other strategy. They indicated the applied strategy by button press after which a blank screen for 2000 ms was presented as inter-trial interval. Participants were carefully instructed on how to report their strategy use at the beginning of the EEG test session following the procedure described by Campbell and Xue (2001). Practice problems were presented to familiarize participants with the experimental task.

EEG was acquired through the BioSemi ActiveTwo system (BioSemi, Amsterdam, The Netherlands) from 64 scalp electrodes placed according to the extended 10–20 systems (see Fig. 2). An electrooculogram (EOG) was recorded from three additional electrodes; two placed horizontally at the outer canthi of both eyes, and one placed above the nasion between in the inner canthi of both eyes. EEG and EOG signals were sampled at 256 Hz and filtered between DC and 128 Hz.

2.3. Procedure

The EEG recording started with a 3 min rest EEG during which the participant was instructed to open the eyes, close them, or to deliberately blink, roll and move

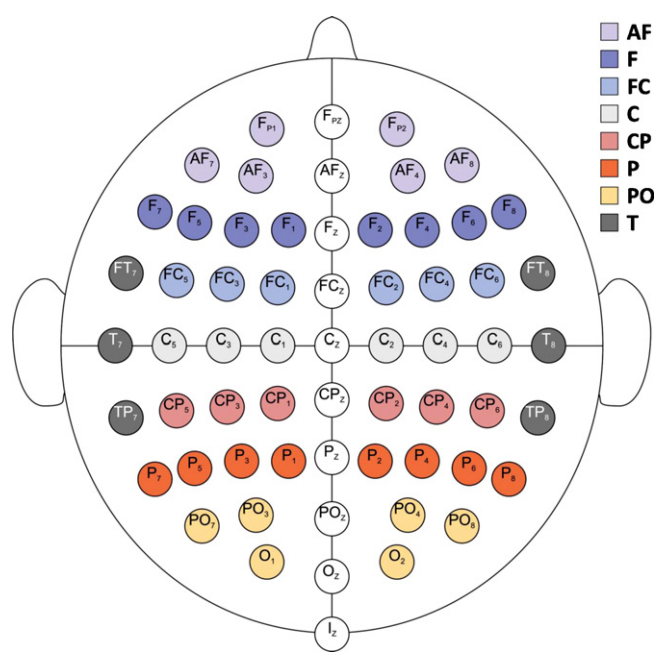

Fig. 2. Schematic display of EEG electrode positions. For statistical analyses, %ERS/ERD was aggregated over 8 areas per hemisphere (electrode positions given exemplarily for the left hemisphere): anteriofrontal (AF; Fp1, AF7, AF3), frontal (F; F7, F5, F3, F1), frontocentral (FC; FC5, FC3, FC1), central (C; C5, C3, C1), centroparietal (CP; CP5, CP3, CP1), parietal (P; P7, P5, P3, P1); parietooccipital (PO; PO7, PO3, O1), and temporal (T; FT7, T7, TP7).

the eyes. This sequence was required for the automatic reduction of eye movement artifacts (see below). Next, participants were presented with the 168 problems in 4 blocks of 42 problems. Each block consisted of only one arithmetic operation, yielding two addition and two subtraction runs. Within each block, the problems were presented in a fixed pseudorandomised order. The addition and subtraction blocks were presented in alternating sequence. Half of the participants started with an addition block, and half of the participants started with a subtraction block. Between blocks, a short break was included to avoid participants losing their concentration. The EEG test session took about 1.5 h including mounting and de-mounting of the electrodes and task instruction.

2.4. Data analysis

Data analysis was similar to the procedure described in De Smedt et al. (2009). EEG data were first band-pass filtered between 0.5 and 45 Hz to eliminate slow-frequency and power-line contamination artifacts. EOG artifacts were automatically reduced by employing a regression method (Schlögl et al., 2007) based on the resting EEG sequence. The continuous EEG data was then divided into trials of 10 s length (3000 ms before and 7000 ms after problem onset), and all trials were visually inspected for artifacts. The spatial information of the artifact-free EEG data was enhanced by applying a 3D spline surface Laplacian estimation (Babiloni et al., 1998). This procedure has turned out to significantly improve the spatial resolution of EEG potential distributions by reducing head volume conductor effects and by cancelling the influence of the electrical reference.

ERS/ERD was computed for correctly solved trials in the theta (3–6 Hz), lower alpha (8–10 Hz) and upper alpha (10–13 Hz) frequency bands (see Klimesch, 1999). The 10 s trial EEG data was bandpass filtered using a Fast Fourier Transformation (FFT) based Finite Impulse Response (FIR) method (Oppenheim and Schafer, 1989), yielding a frequency resolution of 0.1 Hz. Afterwards, each amplitude sample of the filtered data was squared by using a moving window (sample-by-sample) of 500 ms length to obtain power values (μV^2). The data from 500 to 2500 ms after trial onset (during the fixation interval) served as the reference interval (R), and the data from problem presentation (at 3000 ms after trial onset) until 125 ms before the oral response as registered by the voice key was used as activation interval (A) for ERS/ERD computation. The last 125 ms (i.e., 32 samples) of the response latency were discarded to account for the delay of the voice-key trigger signal and to eliminate motor- and speech-related artifacts. For both, R and A intervals, the bandpower values were first averaged over the respective time intervals (horizontal averaging) and then over the trials (vertical averaging), resulting in two values (one for R and one for A) per channel. The amount of ERS/ERD was calculated according to the formula: $\%ERS/ERD = [(A - R)/R] \times 100$. Positive values indicate increases in band power (ERS) and negative values indicate decreases (ERD).

It is important to note that the length of the activation intervals (A) varied between individuals and across trials because these intervals were defined as the time period from problem onset until 125 ms before the response. This analysis procedure has been frequently used in EEG investigations of higher-order cognitive processes (e.g., De Smedt et al., 2009; Grabner et al., 2004; Neubauer et al., 2004) as the activation interval covers the entire time period of problem solving independently of differences between individuals or task conditions. In other words, the variable activation intervals control for differences between individuals, task conditions, and trials, as the bandpower is averaged over time for each trial (and then over trials) before the ERS/ERD is computed. The analysis of fixed activation intervals, in contrast, would neglect such differences and either capture only part of the cognitive processes related to solving the given problem (when the interval length is shorter than the response latency) or additionally include task-unrelated cognitive processes (when interval length is longer than the response latency). For statistical analyses, the %ERS/ERD values were topographically aggregated (by using the arithmetic mean) to obtain 8 cortical areas per Hemisphere (electrode positions given exemplarily for the left hemisphere): anteriofrontal (AF; Fp1, AF7, AF3), frontal (F; F7, F5, F3, F1), frontocentral (FC; FC5, FC3, FC1), central (C; C5, C3, C1), centroparietal (CP; CP5, CP3, CP1), parietal (P; P7, P5, P3, P1); parietooccipital (PO; PO7, PO3, O1), and temporal (T; FT7, T7, TP7) (see Fig. 2).

Behavioral data (accuracy, response latency of the correct trials) were analyzed using repeated measures ANOVAs with Problem Size (small, large) or reported Strategy (retrieval, procedure use) as within-subject factors. For EEG data, similar ANOVAs were computed additionally including Hemisphere (left, right) and Area (8 cortical areas as described above) as within-subject factors. ANOVAs on %ERS/ERD data were conducted separately for the three frequency bands (theta, lower alpha, upper alpha). In all statistical analyses, degrees of freedom were corrected for violations of the sphericity assumption by means of the Huynh–Feldt procedure; the probability of a Type I error was maintained at .05. If applicable, uncorrected df values together with the corrected p value and the Huynh–Feldt epsilon (ε) are reported.

3. Results

The results section is divided into two parts. First, we evaluate the effects of verbal strategy reports and problem size, separately, to assess whether the overall result patterns from both analyses converge. Second, we present the direct comparisons between reported strategies when problem size was held constant and between problem sizes when the reported strategy was held constant in order to investigate the validity of both approaches to capture the neurocognitive processes of arithmetic problem strategies.

3.1. Verbal strategy reports

3.1.1. Behavioral data

Retrieval strategies were reported in 52.01% (SD = 12.71) and procedural strategies in 45.99% (SD = 13.41) of all trials. The percentage of trials on which other strategies were reported was low, 2.00% (SD = 3.40), and these trials were excluded from further analyses. Problems for which retrieval strategy use was indicated were solved more accurately (96.46% (SE = 0.68%) vs. 91.57% (SE = 0.98%); $F(1,18) = 25.84$, $p < .001$, $\eta^2 = .59$) and faster (1103 ms (SE = 47 ms) vs. 2302 ms (SE = 117 ms); $F(1,18) = 160.90$, $p < .001$, $\eta^2 = .90$) than problems for which procedural strategies were reported.

3.1.2. Event-related (de-)synchronisation (%ERS/ERD)

In the theta band, the main effects of Strategy ($F(1,18) = 31.76$, $p < .001$, $\eta^2 = .64$) and Area ($F(7,126) = 9.33$, $p < .001$, $\eta^2 = .34$, $\varepsilon = .30$), and the Strategy × Area ($F(7,126) = 2.91$, $p < .05$, $\eta^2 = .14$, $\varepsilon = .49$) and Strategy × Area × Hemisphere ($F(7,126) = 2.92$, $p < .05$, $\eta^2 = .14$, $\varepsilon = .43$) interactions were significant. As presented in Fig. 3a, reported retrieval use was associated with higher theta ERS than reported procedure use across most of the topographical areas (all $ps < .05$ in post hoc comparisons) except for left anteriofrontal and temporal cortices. The largest difference was observed in the left parietooccipital cortex.

In the lower alpha band, only main effects of Strategy ($F(1,18) = 19.11$, $p < .001$, $\eta^2 = .52$) and Area ($F(7,126) = 2.99$, $p < .05$, $\eta^2 = .14$, $\varepsilon = .48$) emerged (see Online supplementary material Fig. S1a). Reported procedural strategy use was associated with generally higher ERD than reported retrieval use (−18.09% vs. −4.10%).

In the upper alpha band, there were significant Strategy × Area ($F(7,126) = 2.81$, $p < .05$, $\eta^2 = .14$, $\varepsilon = .45$) and Hemisphere × Area ($F(7,126) = 9.04$, $p < .001$, $\eta^2 = .33$, $\varepsilon = .69$) interactions. As depicted in Fig. 3c, the reported procedural compared to retrieval strategy use elicited higher ERD over parietal and parietooccipital cortices (both $ps < .05$). The interaction of Hemisphere by Area showed higher alpha ERD in the right compared to the left parietooccipital region.

3.2. Problem size

3.2.1. Behavioral data

Similar to strategy use, higher accuracies (96.71% (SE = 0.64%) vs. 90.73% (SE = 1.06%); $F(1,18) = 32.60$, $p < .001$, $\eta^2 = .64$) and shorter response latencies (1084 ms (SE = 54 ms) vs. 2085 ms (SE = 110 ms); $F(1,18) = 168.58$, $p < .001$, $\eta^2 = .90$) were found in small compared to large problems.

3.2.2. Event-related (de-)synchronisation (%ERS/ERD)

In the theta band, the ANOVA revealed significant main effects of Problem Size ($F(1,18) = 21.15$, $p < .001$, $\eta^2 = .54$) and Area ($F(7,126) = 9.05$, $p < .001$, $\eta^2 = .34$, $\varepsilon = .28$), as well as interactions of Problem Size × Area ($F(7,126) = 3.12$, $p < .05$, $\eta^2 = .15$, $\varepsilon = .65$) and Problem Size × Area × Hemisphere ($F(7,126) = 3.64$, $p < .01$, $\eta^2 = .17$, $\varepsilon = .64$). The overall pattern of the latter interaction was highly similar to the one observed for strategy use (see Fig. 3b), with small problems eliciting larger theta ERS than large problems. This effect was again significant across most of the topographical areas (all

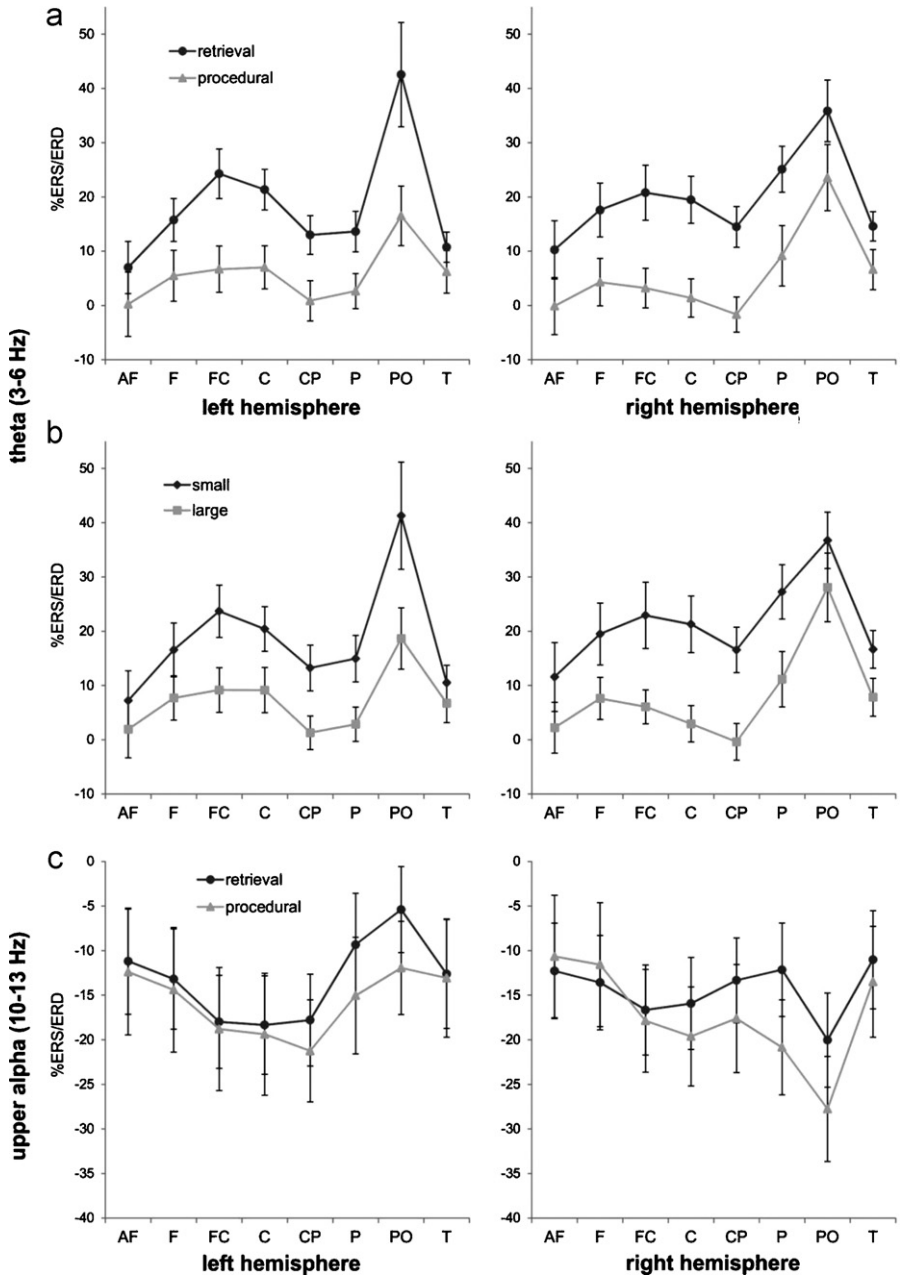

Fig. 3. Impact of strategy use and problem size (when analyzed separately) on event-related (de-)synchronisation (%ERD/ERS). (a) Effect of reported strategy use on %ERD/ERS in the theta band (3–6 Hz). (b) Effect of problem size on %ERD/ERS in the theta band (3–6 Hz). (c) Effect of reported strategy use on %ERD/ERS in the upper alpha band (10–13 Hz). Please note that only the interaction of Strategy and Area reached significance in the upper alpha band. Error bars depict 1 SE of the mean. AF = anteriofrontal, F = frontal, FC = frontocentral, C = central, CP = centroparietal, P = parietal, PO = parietooccipital, T = temporal.

ps < .05) except for left temporal, right parietooccipital and anteriofrontal cortices bilaterally, and was most strongly pronounced in the left parietooccipital areas.

The analysis of ERS/ERD in the lower alpha band only yielded a main effect of Problem Size ($F(1,19) = 25.74$, $p < .001$, $\eta^2 = .59$) but no interactions with this factor (see Online supplementary material Fig. S1b). Solving large problems was accompanied by higher overall alpha ERD (−17.37%) than small problems (−2.28%).

In the upper alpha band, no effect of Problem Size emerged, despite showing a similar topographic distribution (see Online supplementary material Fig. S2). There was only a significant Area × Hemisphere interaction ($F(7,126) = 9.13$, $p < .001$, $\eta^2 = .34$, $\varepsilon = .71$), showing higher alpha ERD in the right compared to the left parietooccipital cortex.

3.2.3. Comparison between the verbal strategy reports and problem size approach

Overall, small problems were predominantly solved with retrieval strategies ($M = 90.86\%$, SD = 11.35%), whereas in 75.71% (SD = 17.56%) of the large problems the use of procedural strategies was reported As less than 10% of the small problems were indicated to be solved by procedural strategies, it was not possible to contrast reported retrieval vs. procedural strategies within the small problem size category and to contrast small vs. large problem sizes within the reported procedural strategy category. Therefore, our direct comparison of the verbal strategy reports and problem size approach is restricted to two analyses. The impact of reported strategy use (retrieval vs. procedure) was evaluated within the large problem size, whereas the effect of problem size (small vs.

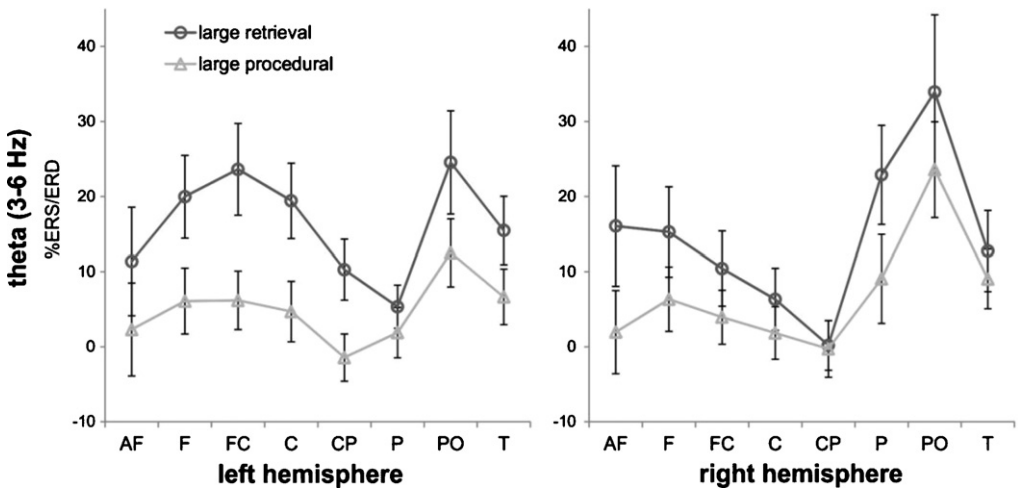

Fig. 4. Effect of reported strategy use on event-related (de-)synchronisation (%ERD/ERS) in the upper alpha band (10–13 Hz) while solving large problems. Error bars depict 1 SE of the mean. AF = anteriofrontal, F = frontal, FC = frontocentral, C = central, CP = centroparietal, P = parietal, PO = parietooccipital, T = temporal.

large) was investigated for those problems that were indicated to be solved by retrieval.

3.2.4. Behavioral data

Large problems for which retrieval strategies were reported were solved faster than those for which procedural strategies were indicated (1304 ms (SE = 62 ms) vs. 2361 ms (SE = 121 ms); $F(1,18) = 134.32$, $p < .001$, $\eta^2 = .88$). Accuracies did not differ significantly between reported retrieval and procedural strategies (93.81% (SE = 1.72%) vs. 91.58% (SE = 1.02%); $p = .20$).

Small retrieval problems were also solved faster than large retrieval problems (1040 ms (SE = 48 ms) vs. 1304 ms (SE = 62 ms); $F(1,16) = 28.31$, $p < .001$, $\eta^2 = .61$) but their accuracies did not differ significantly (97.33% (SE = 0.68%) vs. 93.81% (SE = 1.73%); $p = .07$).

3.2.5. Event-related (de-)synchronisation (%ERS/ERD)

Within the large problems, ERS/ERD in the theta band significantly differed between reported retrieval and procedure use. There was a main effect of Strategy ($F(1,18) = 8.61$, $p < .01$, $\eta^2 = .32$) and a Strategy × Hemisphere × Area interaction ($F(7,126) = 2.20$, $p < .05$, $\eta^2 = .11$, $\varepsilon = .95$). As depicted in Fig. 4, problems reported to be solved by means of retrieval strategies were associated with higher theta ERS, significantly over left frontal, frontocentral, central, centroparietal and temporal as well as over right parietal areas (all *p*s < .05 in post hoc comparisons). In the two alpha bands, no significant effects related to verbal strategy reports were observed (see Online supplementary material Figs. S4a and S5a). Main effects of Area were observed in all three bands (theta: $F(7,126) = 4.50$, $p < .01$, $\eta^2 = .20$, $\varepsilon = .43$; lower alpha: $F(7,126) = 4.09$, $p < .01$, $\eta^2 = .19$, $\varepsilon = .48$; upper alpha: $F(7,126) = 2.91$, $p < .05$, $\eta^2 = .14$, $\varepsilon = .50$), and there were Hemisphere × Area interactions in the theta ($F(7,126) = 4.57$, $p < .01$, $\eta^2 = .20$, $\varepsilon = .53$) and upper alpha bands ($F(7,126) = 4.65$, $p < .01$, $\eta^2 = .21$, $\varepsilon = .52$).

Within the problems that were indicated to be solved by retrieval, no significant effect of Problem Size emerged in any frequency band, indicating that small and large problems did not differ when retrieval strategies were reported (see Online supplementary material Figs. S3, S4b and S5b). The other effects were similar to the analysis within the large problems: main effects of Area in all three bands (theta: $F(7,126) = 4.76$, $p < .01$, $\eta^2 = .21$, $\varepsilon = .39$; lower alpha: $F(7,126) = 3.97$, $p < .05$, $\eta^2 = .18$, $\varepsilon = .43$; upper alpha: $F(7,126) = 3.03$, $p < .05$, $\eta^2 = .14$, $\varepsilon = .50$) and Hemisphere × Area interactions in the theta ($F(7,126) = 3.44$, $p < .05$, $\eta^2 = .16$, $\varepsilon = .48$) and upper alpha bands ($F(7,126) = 4.73$, $p < .01$, $\eta^2 = .21$, $\varepsilon = .53$).

4. Discussion

The present EEG study extends previous research on the oscillatory brain correlates of arithmetic problem solving strategies in two ways. First, we used concurrent, trial-by-trial verbal strategy reports to distinguish between problems that are solved by fact retrieval and those that are solved by procedural strategies. Second, we compared the verbal strategy report approach with the common problem size approach with respect to their association with the neurophysiological data to obtain further insights into their validity for assessing arithmetic strategies. The present findings revealed (a) a general convergence of both approaches in the ERS/ERD data when analyzed separately, and (b) first neurophysiological evidence for a higher validity of verbal strategy reports for capturing solution strategies applied in mental arithmetic.

The separate analyses showed that both, verbal strategy reports and problem size had a strong impact on the ERS/ERD in theta and alpha frequency bands. As hypothesized, we found higher theta ERS for reported retrieval compared to procedural strategies with the largest differences in the left parietooccipital cortex. A comparison on the basis of problem size revealed a strikingly similar pattern, with small problems being accompanied by higher theta ERS than large problems, replicating the results of De Smedt et al. (2009). Oscillations in the theta frequency range have been repeatedly associated with long-term memory encoding and retrieval processes (e.g., Jacobs et al., 2006; Klimesch et al., 2001), and are assumed to be involved in functionally linking the cortex with the medial temporal lobe (in particular the hippocampus; Bastiaansen and Hagoort, 2003; Klimesch et al., 2005). In the language domain, it was proposed that theta activity is particularly sensitive to the retrieval of lexical-semantic information (such as the meaning of words) from long-term memory (for a review, cf. Bastiaansen et al., 2006). The current data, together with those by De Smedt et al. (2009), suggest that theta oscillations may be similarly sensitive to the retrieval of lexical-semantic information in the numerical domain. In mental calculation, lexical-semantic information may consist of arithmetic facts that are assumed to be represented verbally in long-term memory (Dehaene et al., 2003). The retrieval of these facts from long-term memory seems to rely on functional interactions between the medial temporal lobe and (left parietal) cortical structures, which may be mediated by theta oscillations (see also Earle et al., 1996).

With regard to alpha activity, we distinguished a lower and an upper alpha band (Klimesch, 1999). In general, alpha oscillations are assumed to originate from thalamo-cortical and cortico-cortical

networks with their amplitude being inversely related to the activated neuronal population (Klimesch et al., 2007). The lower and the upper alpha frequency band (below and above 10 Hz, respectively) differ in the topographic differentiation and the specificity to cognitive demands (Neuper and Pfurtscheller, 2001; Pfurtscheller and Lopes da Silva, 1999, 2005). ERD in the lower alpha band typically emerges topographically widespread in response to almost any cognitive task and is therefore assumed to indicate basic attentional processes and/or arousal (Fink et al., 2005). We therefore expected strategy use and problem size to affect the overall level of lower alpha ERD, which was indeed the case: procedural strategies and large problems elicited higher overall ERD than retrieval strategies and small problems, suggesting stronger demands on arousal and attention during procedural strategy use and while solving large problems.

In the upper alpha band, ERD usually emerges over topographically restricted (task-relevant) areas and is regarded to reflect more specific task requirements (Pfurtscheller and Lopes da Silva, 2005). The specificity of these task requirements, however, is still a matter of discussion (Klimesch et al., 2006). Given that also upper alpha ERD emerges in various cognitive tasks, it has been argued that it can be regarded as correlate of cortical activation during cognitive information processing (Neubauer et al., 2006). Against this background, upper alpha ERD was hypothesized to be higher during procedural than during retrieval strategies over task-relevant parietooccipital regions. In fact, this ERD pattern was observed. This finding suggests stronger cortical activation of bilateral parietooccipital networks during the application of procedures than during fact retrieval and is in line with fMRI studies showing more pronounced bilateral parietal brain activity in task conditions that more strongly rely on procedural strategy use, such as complex (vs. simple) problems (e.g., Kong et al., 2005) and untrained (vs. trained) problems (e.g., Delazer et al., 2003). Moreover, it corresponds to the findings by Grabner et al. (2009a) who used post-scan strategy reports. Unlike self-reported strategy use, problem size had no impact on the ERD in the upper alpha band. This result stands in contrast to De Smedt et al. (2009) and may be partly due to differences in problem selection and EEG analysis. First, the large problems in the latter study involved larger numbers which may have elicited more procedural strategy use than the large problems in the current study. Second, while De Smedt et al. (2009) focused on one alpha band including lower and upper alpha frequencies, we distinguished two alpha bands based on their differential functional significance.

Taken together, the separate analyses of both approaches (verbal strategy reports vs. problem size) yielded very similar neurophysiological results. In the theta band, the topographic differences between self-reported retrieval and procedural strategy use were practically indistinguishable from those between small and large problems. In the lower alpha band, both approaches had a comparably large impact on overall ERS/ERD. In the upper alpha band, however, brain responses were modulated only by the classification based on verbal reports but not by problem size, already suggesting that the first approach could be somewhat better suited to capture the arithmetic strategies actually applied.

It should be noted that there was a high overlap between the classification based on reported strategy use and that based on problem size. Nearly all small problems were indicated to be retrieved from memory (about 91%) and the majority of the large problems were solved by procedural strategies (about 76%). This overlap may in itself explain the high neurophysiological convergence of both approaches. We therefore conducted additional analyses in which we directly compared both approaches. More specifically, we compared the data of the retrieval and procedure trials within the large problem size, on the one hand, and the data of the small and large trials for which retrieval was reported, on the other hand.

The comparison of both approaches provided further and direct evidence for a higher sensitivity of the verbal strategy report approach. It turned out that self-reported strategy use further explained variance within a category of problem size (i.e., in large problems), whereas problem size within a category of strategy (i.e., in retrieval problems) did not. More specifically, we found higher theta ERS for large retrieval compared to large procedural problems, which was most strongly pronounced in the left hemisphere over frontal to centroparietal areas. In contrast, problem size did not modulate ERS/ERD in any frequency band within problems that were indicated to be solved using the same strategy (i.e., retrieval). These neurophysiological data suggest that verbal strategy reports are more closely linked to the underlying neurocognitive processes involved in arithmetic strategies than the distinction based on problem size.

These findings have important implications for the interpretation of studies on the neural correlates of arithmetic strategies that only rely on the problem size method, i.e., those contrasting the brain responses for small vs. large problems to assess retrieval vs. non-retrieval strategies (e.g., De Smedt et al., 2009; Grabner et al., 2007a; Jost et al., 2004a,b; Kong et al., 2005; Ku et al., 2010; Nunez-Pena et al., 2006; Stanescu-Cosson et al., 2000; Zago et al., 2001). In particular, the present data suggest that the large problem size category might contain both retrieval and non-retrieval strategies. Our findings also add to the controversial discussion on the validity of verbal strategy reports in general (Kirk and Ashcraft, 2001; Smith-Chant and LeFevre, 2003). For example, it has been argued that participants may change their solution strategies when asked to describe them, that they may be unable to report them accurately, or that specific experimental conditions (e.g., emphasizing speed or accuracy) bias solution procedures and verbal reports. The present data provide first neurophysiological evidence that verbal strategy reports are more strongly associated with the underlying neurocognitive processes than the problem size approach. In other words, verbal strategy reports appear to capture a larger amount of the variability in actual strategy use than the method of experimentally manipulating the size of the presented problems.

At this point, two important limitations of the present study need to be noted. First, due to the fact that small problems are almost exclusively solved by fact retrieval in adults (e.g., Campbell and Xue, 2001), it was not possible to investigate the impact of strategy use in small problems or the impact of problem size in procedural problems. Consequently, the present comparative results of both approaches are based on large problems (impact of strategy use) and on retrieval problems (impact of problem size). An investigation of reported strategy use and problem size as largely independent factors would require a study design in which also the reported strategy is experimentally manipulated, for example by the administration of an arithmetic training such as in Grabner et al. (2009b).

Second, we focused on the time interval from problem onset until 125 ms before the response for calculating the ERS/ERD values. These values are thus based on activation intervals of different lengths as small and retrieval problems are solved faster than large and procedural problems, respectively. The current results may therefore not be directly comparable to EEG studies that have analyzed fixed time intervals. However, activation intervals of different lengths are often used in ERS/ERD research, and this method can be regarded to be superior to fixed time intervals in capturing all task-related cognitive processes, particularly in higher-order cognitive tasks where there are notable differences between subjects and task conditions (e.g., De Smedt et al., 2009; Grabner et al., 2004; Neubauer et al., 2004). A fixed activation interval might either capture only part of the cognitive processes related

to problem solving (when the interval length is shorter than the response time) or additionally include task-unrelated cognitive processes such as rest (when interval length is longer than the response time).

Our finding of strategy differences within problems of similar size in the theta band adds further evidence to the current notion that this frequency range may be particularly sensitive to the retrieval of semantic information from long-term memory. Theta oscillations have also been related to working memory functions, including attentional and executive processes (e.g., Gevins et al., 1997; Jensen and Tesche, 2002; Kahana et al., 2001; Sauseng et al., 2007; Smith et al., 1999), and in some studies on number processing, theta power changes have been attributed to these working memory processes rather than to information retrieval from long-term memory (e.g., Harmony et al., 1999; Moeller et al., 2010). However, theta effects in working memory tasks (a) typically show higher theta power (or ERS) with increasing working memory load or attentional demands and (b) are mainly restricted to the (mid-)frontal cortex. In contrast to this result pattern, we found higher theta ERS in the easier task conditions, that is in retrieved and in small problems, with the largest difference in the parietooccipital cortex. Thus, the present findings cannot be reconciled with this alternative (working memory) interpretation.

5. Conclusion

The analysis of oscillatory EEG activity based on verbal strategy reports yielded results that were largely similar to the analysis following the common problem size approach. Most importantly, a direct comparison of the verbal strategy report and the problem size approach only revealed significant differences in ERS/ERD between reported strategies within the same problem size but not between problem sizes within the same reported strategy. This finding indicates that verbal strategy reports are more closely linked to neurophysiological data than the problem size approach. Taken together, the findings provide first neurophysiological evidence for the validity of verbal strategy reports to capture strategies in mental arithmetic.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biopsycho.2011.02.019.

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